DATA SCIENCE ROADMAP

From Complete Beginner to Industry Expert

Mathematics • Python • Statistics • Excel • Power BI • Machine Learning • Deep Learning • MLOps

COMPLETE ROADMAP

What Is Data Science?

Data Science is the art and science of transforming raw data into meaningful insights, decisions, and products. It sits at the crossroads of mathematics, computer programming, domain expertise, and storytelling. A data scientist collects messy, real-world data, cleans and organises it, explores patterns hidden within it, builds predictive models using statistics and machine learning, and communicates findings in a way that drives real business or research value.

The Three Pillars of Data Science

Pillar	What It Means	Example Skills
Mathematics & Statistics	The theoretical backbone that makes predictions trustworthy and valid.	Probability, Linear Algebra, Hypothesis Testing
Programming & Engineering	Tools to process millions of rows of data quickly and efficiently.	Python, SQL, Spark, Cloud Platforms
Domain Knowledge	Understanding the business or field so you ask the right questions.	Finance, Healthcare, Marketing, Retail

Why Data Science Is One of the Best Careers Today

- High demand across almost every industry worldwide
- Excellent salaries among top-paying tech roles globally
- Remote-friendly work culture in most companies
- Creative problem-solving every single day
- Constant learning — the field never gets boring
- Measurable impact — your models affect business outcomes directly
- Diverse paths: analytics, AI research, engineering, consulting
- Strong global community and open-source culture

> **"Without data you are just another person with an opinion." — W. Edwards Deming**

Your End-to-End Learning Roadmap

This guide is structured into 7 progressive phases. Each phase builds on the previous. Do not skip phases — the foundation always matters more than the fancy tools on top. The total duration is approximately 12 to 18 months of consistent study.

#	Phase	Key Outcome
1	Foundation & Mindset	Understand the field, tools, and study habits
2	Mathematics & Statistics	Solid mathematical backbone for all ML models
3	Excel & Data Tools	Analyse data with spreadsheets and pivot tables
4	Programming with Python & SQL	Write code to load, clean, and explore data
5	Data Visualisation & BI Tools	Build dashboards in Power BI and Python
6	Machine Learning & Deep Learning	Build, train, and evaluate predictive models
7	Advanced Topics & Career Launch	MLOps, specialisations, portfolio, and jobs

PHASE 1 Foundation & Mindset

Duration: 2–3 Weeks

What to Set Up First

Before writing a single line of code, set up your environment and understand the landscape. Many beginners skip this and waste months going in wrong directions.

Tool / Action	Why It Matters	Where to Get It
Anaconda (Python distribution)	Installs Python + 100+ libraries in one go	anaconda.com
Jupyter Notebook / VS Code	Interactive coding environment for learning	code.visualstudio.com
Git & GitHub account	Version control — saves your code history	github.com
Kaggle account	Free datasets, notebooks, and competitions	kaggle.com
Google Colab	Free GPU in the browser — no local setup needed	colab.research.google.com

Core Concepts to Understand This Week

- What structured vs unstructured data means
- The difference between AI, ML, and Data Science
- What a dataset, row, column, and variable are
- Types: numerical, categorical, ordinal, text, image
- What a model is and what it tries to do
- Supervised vs unsupervised learning (just the idea)
- What overfitting means in plain English
- Why data quality matters more than algorithm choice

PHASE 2 Mathematics & Statistics

Duration: 6–8 Weeks | The Most Important Phase

You do not need a university degree in mathematics, but you do need to understand the core ideas deeply enough to know why algorithms behave the way they do. Skipping this phase produces data scientists who can copy-paste models but cannot debug them or explain their results.

2A — Linear Algebra

Linear algebra is the language of data. Every dataset is a matrix. Every transformation — scaling, rotating, projecting — is a matrix operation. Neural networks are stacks of matrix multiplications.

- Scalars, vectors, matrices, and tensors
- Matrix addition and multiplication
- Dot product and its geometric meaning
- Transpose and inverse of a matrix
- Eigenvalues and eigenvectors (key for PCA)
- Norms — L1 (Manhattan) and L2 (Euclidean)
- Singular Value Decomposition (SVD)
- Practical tool: NumPy arrays in Python

2B — Calculus

Gradient descent — the engine behind every neural network — is pure calculus. You need derivatives to understand how a model learns by minimising its error function step by step.

- Functions, limits, and continuity
- Derivatives and the chain rule
- Partial derivatives for multivariate functions
- Gradient — direction of steepest increase
- Gradient descent — intuition and mathematics
- Learning rate and its effect on convergence
- Backpropagation (chain rule in neural networks)
- Practical tool: SymPy for symbolic math

2C — Probability & Statistics

Statistics lets you make confident claims from uncertain data. Probability tells you how likely something is. Together they are the scientific foundation of every model you build.

Topic	Key Concepts	Used In
Descriptive Statistics	Mean, median, mode, variance, standard deviation, IQR, skewness	EDA, feature engineering
Probability Theory	Events, sample space, conditional probability, Bayes theorem	Naive Bayes, probabilistic models
Distributions	Normal, Binomial, Poisson, Uniform, Exponential, Beta	Regression assumptions, simulations

Topic	Key Concepts	Used In
Inferential Statistics	Hypothesis testing, p-value, confidence intervals, Type I/II errors	A/B testing, model validation
Correlation & Regression	Pearson/Spearman correlation, simple & multiple linear regression	Prediction, feature selection

PHASE 3 Excel & Spreadsheet Skills

Duration: 2–3 Weeks | Underrated but Essential

Excel is used in every company in the world. Before writing Python code, being able to open a spreadsheet, understand its structure, and perform analysis is a baseline professional skill. Many real data science tasks begin and end in Excel.

Skill Area	What to Learn
Basics	Cells, ranges, rows, columns, formatting, named ranges, data types
Formulas	SUM, AVERAGE, IF, VLOOKUP, HLOOKUP, INDEX-MATCH, SUMIF, COUNTIF
Data Cleaning	Remove duplicates, text-to-columns, find & replace, data validation, flash fill
Pivot Tables	Group, summarise, filter, slice, and cross-tabulate large datasets quickly
Charts & Graphs	Bar, line, scatter, pie, histogram, combo charts — choose the right one
Statistical Tools	Excel's Data Analysis ToolPak — regression, descriptive stats, t-tests
Power Query	Import, transform, and merge data from multiple sources without code
Keyboard Shortcuts	Ctrl+Shift+L (filters), Ctrl+T (tables), Alt+= (AutoSUM), F4 (lock cell reference)

Excel Practice Projects

- Sales dashboard — import sales CSV, pivot by region, chart monthly trends.
- Student score analyser — calculate class average, top and bottom 10%, pass/fail rates.
- Budget tracker — income vs expense categories, monthly summary, forecast next 3 months.
- HR attrition report — use VLOOKUP and pivot to analyse why employees leave.

PHASE 4 Python & SQL for Data Science

Duration: 6–8 Weeks

Python is the primary language of data science. SQL is the language every data professional must speak to fetch and manipulate data from databases. Together they handle 90% of real-world data work.

4A — Python Fundamentals

Topic	Concepts to Cover
Core Python	Variables, data types, loops, conditionals, functions, list comprehensions
Data Structures	Lists, tuples, dictionaries, sets — when to use each and why
File Handling	Read/write CSV, JSON, and text files with Python's built-in open()
Error Handling	try/except blocks, raising exceptions, writing robust scripts
OOP Basics	Classes, objects, methods, inheritance — enough to read library source code

4B — Python Libraries for Data Science

Library	Purpose	Key Functions to Learn
NumPy	Fast numerical arrays	array, reshape, dot, broadcast, arange, linspace
Pandas	Data manipulation	read_csv, DataFrame, groupby, merge, pivot_table, fillna, dropna
Matplotlib	Basic plotting	plot, scatter, bar, hist, subplots, xlabel, title, savefig
Seaborn	Statistical visualisation	heatmap, pairplot, boxplot, violinplot, distplot
Scikit-learn	Machine learning	train_test_split, StandardScaler, Pipeline, GridSearchCV
SciPy	Scientific computing	stats module — t-tests, ANOVA, curve fitting
Statsmodels	Statistical models	OLS regression, logit, time series decomposition

4C — SQL for Data Science

In most companies, data lives in relational databases — MySQL, PostgreSQL, BigQuery, Snowflake. SQL is not optional. Learn to write queries that extract exactly the data you need.

SQL Level	Topics Covered
Beginner	SELECT, FROM, WHERE, ORDER BY, LIMIT, DISTINCT, basic data types
Intermediate	GROUP BY, HAVING, JOIN (INNER, LEFT, RIGHT, FULL), subqueries, aliases

SQL Level	Topics Covered
Advanced	Window functions (ROW_NUMBER, RANK, LAG, LEAD), CTEs (WITH clause), CASE WHEN
Data Science	Aggregate queries, date functions, string manipulation, writing analytical queries

Phase 4 Projects to Build

- Exploratory Data Analysis (EDA) on Titanic dataset using Pandas and Seaborn.
- COVID-19 data analysis — download WHO data, clean it, and plot country comparisons.
- SQL sales analysis — write 20+ queries on a sample e-commerce database.
- IPL cricket stats analyser — merge multiple CSVs, answer 15 business questions.

PHASE 5 Data Visualisation & Business Intelligence

Duration: 3–4 Weeks

Analysis means nothing if you cannot communicate it clearly. Visualisation turns numbers into stories. Business Intelligence tools like Power BI allow non-technical stakeholders to explore data themselves through interactive dashboards without needing any coding knowledge.

Power BI — The Industry Standard BI Tool

Module	What to Learn
Connecting Data	Import from Excel, CSV, SQL databases, web, Azure, SharePoint
Power Query (M)	Transform data: filter rows, split columns, merge tables, unpivot
DAX Language	Calculated columns, measures, CALCULATE, FILTER, ALL, SUMX, DIVIDE
Report Design	Choose the right visual, colour theory, layout, and accessibility principles
Visuals	Bar/column, line, card, KPI, map, matrix, treemap, custom visuals
Interactivity	Slicers, filters, cross-filtering, drillthrough, bookmarks, tooltips
Publishing	Publish to Power BI Service, share reports, set up data refresh schedules

Visualisation Best Practices

- Choose bar charts for comparisons, line charts for trends
- Use a maximum of 2 to 3 colours per dashboard
- Always label axes and include units of measurement
- Start with the insight, not the chart type
- Use colour to highlight, not just to decorate slides
- Keep your audience in mind — executive vs analyst view

- Avoid 3D charts — they always distort perception
- White space is your friend — do not crowd dashboards

Python Visualisation Stack

- Matplotlib — low-level, fully customisable plots
- Altair — declarative charting for quick exploration
- Seaborn — statistical plots with beautiful defaults
- Folium — geographical maps and choropleths
- Plotly — interactive web-ready charts
- Bokeh — interactive dashboards in the browser

Power BI Dashboard Projects

- Sales Performance Dashboard — total revenue, top products, regional breakdown, monthly trend.
- HR Analytics Report — headcount, attrition rate, department breakdown, salary bands.
- Financial P&L Dashboard — income statement, expense categories, year-over-year comparison.

PHASE 6 Machine Learning & Deep Learning

Duration: 10–14 Weeks | The Core of Data Science

Machine Learning is the process of teaching a computer to learn patterns from data without explicitly programming every rule. Deep Learning is a subfield that uses neural networks inspired by the human brain. This is the phase most people think of when they say 'data science'.

6A — Classical Machine Learning Algorithms

Algorithm	Type	When to Use It
Linear Regression	Supervised	Predict continuous values — house prices, temperature, sales
Logistic Regression	Supervised	Classify into two categories — spam/not spam, fraud/legitimate
Decision Trees	Supervised	Interpretable classification or regression, easy to explain
Random Forests	Supervised	High accuracy, handles missing values, feature importance
Gradient Boosting (XGBoost, LightGBM)	Supervised	Win Kaggle competitions, tabular data king
Support Vector Machines	Supervised	High-dimensional data, image classification baseline

Algorithm	Type	When to Use It
K-Nearest Neighbours	Supervised	Simple classification, recommendation system baseline
K-Means Clustering	Unsupervised	Customer segmentation, image compression
DBSCAN	Unsupervised	Density-based clustering, outlier detection
PCA	Unsupervised	Dimensionality reduction, visualising high-dimensional data
Naive Bayes	Supervised	Text classification, spam detection, fast baseline

6B — The Machine Learning Workflow

Step	Action
1. Define the Problem	What are you predicting? What is success? What data do you have?
2. Collect Data	Gather from databases, APIs, web scraping, public datasets, sensors
3. Explore (EDA)	Understand distributions, correlations, outliers, missing values
4. Clean & Preprocess	Handle nulls, encode categoricals, scale features, remove duplicates
5. Feature Engineering	Create new meaningful features from existing raw columns
6. Select & Train Model	Split data 80/20, fit model on training set, tune hyperparameters
7. Evaluate	Use right metrics — accuracy, F1, AUC-ROC, RMSE, MAE
8. Deploy	Serve predictions via API or integrate into business workflow
9. Monitor	Track model performance over time, retrain when drift occurs

6C — Model Evaluation Metrics

- Accuracy — % of correct predictions (misleading on imbalanced data)
- Precision — of all positive predictions, how many are correct
- Recall (Sensitivity) — of all actual positives, how many did we catch
- F1 Score — harmonic mean of precision and recall
- AUC-ROC — model's ability to separate classes across thresholds
- MAE — average absolute prediction error (regression)
- MSE / RMSE — penalises large errors more heavily
- R-squared — proportion of variance explained by model
- Confusion Matrix — full breakdown of TP, TN, FP, FN
- Cross-validation — k-fold for reliable performance estimates

6D — Deep Learning

Deep Learning uses multi-layered artificial neural networks to learn representations of data. It powers image recognition, natural language processing, speech synthesis, and generative AI.

Architecture	What It Does	Applications
Feedforward NN (MLP)	General-purpose learner	Tabular data, simple classification tasks
CNN	Learns spatial features via filters	Image recognition, video analysis, OCR
RNN / LSTM	Processes sequential data with memory	Time series, text generation, speech
Transformer	Self-attention over full sequences	BERT, GPT, translation, summarisation
Autoencoder	Compress and reconstruct data	Anomaly detection, dimensionality reduction
GAN	Generator vs Discriminator game	Image synthesis, data augmentation

Phase 6 Projects to Build

Project	Algorithms Used	Skills Practised
House Price Prediction	Linear Regression, XGBoost	Regression, feature engineering
Customer Churn Prediction	Logistic Regression, Random Forest	Classification, imbalanced data
Sentiment Analysis (movie reviews)	Naive Bayes, TF-IDF, LSTM	NLP, text preprocessing
Image Classifier (cats vs dogs)	CNN with Keras	Deep learning, data augmentation
Fraud Detection	Isolation Forest, XGBoost	Anomaly detection, class imbalance
Customer Segmentation	K-Means, PCA	Clustering, visualisation
Time Series Forecasting	ARIMA, LSTM	Temporal data, seasonality
Recommender System	Collaborative filtering, Matrix Factorisation	Unsupervised, user-item matrices

PHASE 7 Advanced Topics & Career Launch

Duration: 8–12 Weeks | Become Expert + Employable

7A — MLOps (Machine Learning Operations)

Building a model on your laptop is step one. Getting it to run reliably in production for thousands of users is a completely different challenge. MLOps bridges the gap between data science and software engineering.

- Docker — containerise your model and all dependencies
- GitHub Actions — automate testing and deployment (CI/CD)

- Flask / FastAPI — wrap your model in a REST API
- MLflow — track experiments, parameters, and metrics
- DVC — version control for large datasets
- AWS SageMaker / GCP Vertex AI / Azure ML — cloud platforms
- Apache Airflow — schedule and orchestrate data pipelines
- Evidently AI — monitor model drift in production

7B — Specialisation Tracks

After mastering the fundamentals, pick one area to go deep. Specialists earn more and are hired faster than generalists who know a little of everything.

Specialisation	Key Skills	Typical Job Title
NLP / LLMs	BERT, GPT, RAG, prompt engineering, fine-tuning	NLP Engineer, AI Engineer
Computer Vision	CNN, YOLO, OpenCV, image segmentation, OCR	CV Engineer, Robotics AI
Time Series / Finance	ARIMA, GARCH, Prophet, portfolio optimisation	Quant Analyst, Risk Analyst
Recommender Systems	Matrix factorisation, collaborative filtering, A/B	ML Engineer, Product Data Scientist
Data Engineering	Spark, Kafka, Airflow, dbt, data warehouses	Data Engineer, Analytics Engineer
AI Research	Reading papers, reproducing SOTA, publishing results	Research Scientist

7C — Big Data & Cloud Tools

- Apache Spark — process terabytes of data in parallel
- Hadoop — distributed storage and batch processing
- Kafka — real-time data streaming at massive scale
- dbt — transform data inside your warehouse with SQL
- AWS (S3, Redshift, SageMaker, Lambda, EMR)
- Google Cloud (BigQuery, Vertex AI, Dataflow)
- Azure (Synapse, Azure ML, Databricks)
- Snowflake — cloud-native data warehouse solution

Free Platforms to Learn & Practice

Platform	Best For
Kaggle	Datasets, competitions, free GPU notebooks, community discussion
Google Colab	Free Jupyter-like notebooks with GPU and TPU access
LeetCode (SQL)	Practice SQL interview questions — aim for 50 medium problems
Papers with Code	Read and reproduce recent state-of-the-art ML research papers

Getting Hired — Job Search Strategy

Job Roles in Data Science

Role	Primary Responsibilities	Skills Emphasis
Data Analyst	Dashboards, reports, SQL queries, business insights	SQL, Excel, Power BI, communication
BI Analyst	Self-serve analytics, stakeholder reporting	Power BI, Tableau, DAX, business acumen
Data Scientist	Build and deploy ML models, run experiments	Python, ML, statistics, storytelling
ML Engineer	Scale and productionise models, build ML pipelines	Python, MLOps, Docker, cloud platforms
Data Engineer	Build and maintain data infrastructure pipelines	SQL, Spark, Airflow, cloud, dbt
AI / NLP Engineer	LLMs, fine-tuning, RAG pipelines, chatbots	PyTorch, Hugging Face, prompt engineering
Research Scientist	Publish papers, advance state of the art in AI	Deep math, research writing, experimentation

Interview Preparation

- Statistics & Probability: Be ready for 5 to 10 concept questions
- SQL: Practice window functions, GROUP BY, complex JOINs daily
- Python coding: LeetCode Easy/Medium problems in Python
- ML concepts: Explain every algorithm from intuition to math
- Case studies: Practise end-to-end how-would-you-approach-X questions
- System design: How would you build a recommendation system at scale?
- Behavioural: STAR method for 10 project stories
- Mock interviews: Use Pramp, Interviewing.io, or peer practice

Complete Learning Path — Quick Reference

Phase	Topics	Tools
1. Foundation	What is DS, environment setup, mindset	Anaconda, Jupyter, VS Code, Git, Kaggle
2. Mathematics	Linear algebra, calculus, probability, statistics	NumPy, SymPy, Khan Academy, 3Blue1Brown
3. Excel	Formulas, pivot tables, Power Query, charts	Microsoft Excel, Google Sheets
4. Python & SQL	Core Python, Pandas, NumPy, Seaborn, SQL queries	Python 3, Jupyter, PostgreSQL, SQLite
5. Visualisation & BI	Charts, storytelling, dashboards, DAX	Power BI, Tableau, Plotly, Matplotlib
6. ML & Deep Learning	Algorithms, workflow, evaluation, neural networks	Scikit-learn, XGBoost, TensorFlow, PyTorch
7. Advanced & Career	MLOps, specialisation, portfolio, interviews	Docker, FastAPI, MLflow, AWS/GCP, GitHub

Data science is one of the most rewarding and impactful careers of our generation. Every week you study brings you one step closer to solving real problems with data — whether that is helping doctors detect cancer earlier, helping banks catch fraud, helping farmers predict drought, or helping companies serve their customers better.